

UI/UX Design Specification

Offline Field Issue Tracker

1. Experience Principles

- 1. **Local truth is visible:** “Saved on this device” is never confused with “Synced to server”.
- 2. **One primary action:** the field form emphasizes Save Draft or Submit, not navigation.
- 3. **Recoverable by default:** every failure states what happened and what the user can do.
- 4. **Readable in difficult conditions:** high contrast, large type, strong borders, and no color-only meaning.
- 5. **Progressive detail:** workers see essential fields first; coordinators see dense operational detail.

2. Responsive Surfaces

Surface	Target	Layout
Field capture	360–768 px mobile	Single column, sticky bottom action bar
Queue manager	Mobile/tablet	Cards with status and Retry
Coordinator dashboard	≥1024 px desktop	Filter rail, table/list, detail drawer

3. Connectivity Status Banner

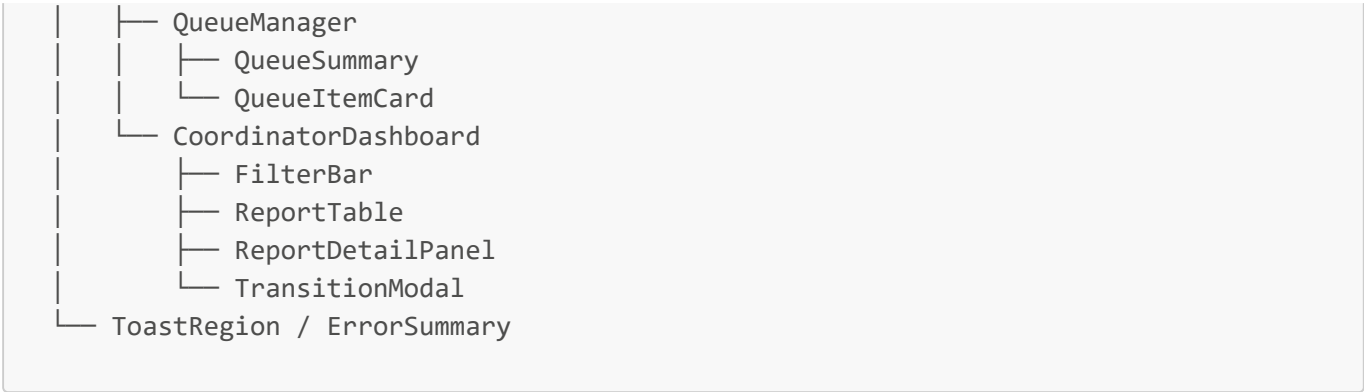
The banner is persistent at the top of the app and includes text plus iconography:

State	Copy	Behavior
Online	Online	Normal background; sync worker may run
Offline	Offline – saved reports will sync later	Amber/gray; disable no local actions
Syncing	Syncing 2 of 5	Progress indicator; do not hide queue
Sync Error	Sync error – Review queue	Red/amber; links to failed item details

The banner must not claim online reachability solely from `navigator.onLine`; a failed request can move the app to `Sync Error`.

4. Component Hierarchy and States





Status badges always pair a label with an icon and pattern:

Status	Meaning	Example text
Draft	Not submitted	Draft
Pending Sync	Durable locally, awaiting server	Pending sync
Synced	Server acknowledged	Synced
Failed	Action required	Failed – Fix and retry

5. Wireframe Flows

5.1 Field Worker Report Form

[≡] New issue[Offline]

Category *
[Select infrastructure type v]

Description *
[What did you observe?]
[]

Priority
[Low] [Medium] [High]

Location
[Use current GPS location]
GPS: Waiting / Captured / Denied
[Latitude] [Longitude]
[Manual landmark / instructions]
Source: GPS / Manual

[Save draft] [Submit report]

Flow:

- 1. Request GPS only after the worker taps the GPS action.
- 2. Display accuracy and capture time when a fix is available.
- 3. If denied or unavailable, keep the form usable and focus manual fields.
- 4. Commit locally before showing the resulting status.
- 5. Navigate to the queue with a confirmation such as **Report saved locally**.

5.2 Queue Manager

Report queue
3 pending · 1 failed · [Sync now]

[!] FAILED
Water leak · High
“Description is required”
[Open] [Retry after fixing]

[⌛] PENDING SYNC
Damaged barrier · Medium
Saved 10:42 · Retry 2
[Retry]

Manual Retry is always explicit, disabled while the item is actively syncing, and reports its result inline.

5.3 Coordinator Review and Transition

Dashboard: [Status v] [Priority v] [Category v] [Search]

ID	Summary	Status	Priority	Updated
...	Road damage	Submitted	High	10:42

Transition modal

Move “Road damage”
Current: Submitted
New status: [Assigned v]
Assignee: [Select v]
Note: [Optional/required]
[Cancel] [Confirm]

For **Rejected**, the note is required and the modal states that rejection is permanent for this report unless a follow-up is created. A 409 refreshes the detail panel and explains that another user changed the report.

6. Accessibility and Field Ergonomics

- Minimum touch target: 48×48 CSS px with 8 px spacing.
- Text and controls target WCAG AA contrast; do not use red/green alone.
- Visible focus ring and full keyboard operation for desktop.
- Labels are persistent, not placeholder-only.
- Error summary appears before the form fields and each invalid field has an inline message.
- Support zoom to 200% without loss of core functionality.
- Use `aria-live="polite"` for sync updates and `aria-live="assertive"` only for blocking errors.
- Avoid motion as the sole progress signal; use text and determinate counts.
- Keep primary actions within thumb reach on mobile.
- Use large line height and concise instructions suitable for low visibility.
- Preserve entered text after validation failure or network failure.

7. Content and Interaction Rules

Use plain language: `Saved on this device`, not `IndexedDB write succeeded`. Never say `Submitted` when only a local transaction completed. Destructive or irreversible actions require a confirmation that names the report and consequence.